

Dylan Slye

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EDUCATION

University of Central Florida

B.S. Data Science, College of Sciences — GPA 3.602 / 4.00

Orlando, FL

Expected December 2027

Minors in Mathematics and Statistics

Awards: Dean's List, UCF · Perfect Math ACT (36) · AP Scholar with Distinction · National Merit Commended Scholar

Coursework in Progress (Fall 2026): Data Science I, Programming for Data Science and AI, Statistics for AI I, Big Data Analytics, Differential Equations

Completed Coursework: Statistics, Statistical Methods, Linear Algebra, Numerical Computing, Computer Science I, Discrete Mathematics, Calculus I–III

TECHNICAL SKILLS

Languages: Python, C (proficient); Java, JavaScript, TypeScript, C++ (familiar)

Systems & Tools: Git, PostgreSQL, Node.js, Next.js, Selenium, REST APIs, job queues, Claude API

ML & Data: NumPy, pandas, scikit-learn, XGBoost, neural networks implemented from scratch (backpropagation, Adam), gradient boosting, logistic regression, feature engineering, model evaluation, data pipeline design

PROJECTS

NFL Pre-Snap Play-Calling Prediction

2026 – Present

Independent Research — Python, NumPy, pandas, XGBoost, scikit-learn

- Implemented logistic regression and a two-hidden-layer neural network from scratch in NumPy, including backpropagation, mini-batch Adam optimization, and early stopping; benchmarked random forests and XGBoost.
- Built an end-to-end run/pass prediction pipeline over 343,000+ curated NFL offensive snaps from 2016–2025, restricting model inputs to information observable before the snap.
- Engineered pre-snap features covering game situation, offensive personnel and formation, team identity, and lagged play-calling tendencies; ran chronology and perturbation audits to detect target leakage.
- Designed a weekly expanding-window backtest that retrained on the previous season plus completed current-season weeks while holding the entire target week out of sample.
- Selected a regularized depth-7 global XGBoost model on 2022–2023 validation, reaching **76.5% chronological out-of-sample accuracy** across 262,953 plays from 2018–2025, roughly 15.5 points above the majority-class baseline, including 76.4% on the most recent full season.
- Rejected a specialized expert-blend ensemble that beat the global model by only 172 net corrections, roughly one extra correct call per 1,529 plays, as unjustified complexity.

ReelForge — Automated Video Generation Engine

2025 – 2026

TypeScript, PostgreSQL, multi-API orchestration

- Designed and shipped a TypeScript pipeline generating documentary-style video end to end, orchestrating Claude API scripting, Fish Audio TTS, Replicate image generation, and Remotion rendering.
- Built the orchestration layer as a PostgreSQL-backed state machine on a pg-boss job queue with a per-run cost ledger, holding generation cost under \$0.50 per video.
- Reached consistent, repeatable output after iterative quality-control tuning; ended active development when quality plateaued below the bar I had set for the format.

EXPERIENCE

Slye Marketing LLC — Software Developer (Contract)

Fall 2025 – Present

Remote

- Sole developer building Python automation and internal tooling across client engagements; owned architecture, implementation, testing, and deployment on concurrent projects.

LEADERSHIP

CCPS Youth Coding — Co-Founder & Lead Instructor

Fall 2023 – Spring 2025

Collier County, FL

- Co-founded a volunteer computer science program reaching 150+ middle school students across nine weekly sections at two schools; wrote the Python curriculum, trained student instructors, and built a succession pipeline that kept it running after we left.